

DG INTERNS HUB

CYBERSECURITY INTERNSHIP

WEEK 2 ASSIGNMENT REPORT

Advanced Networking, Analysis
& Security Concepts

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Week: 2 Tools Used: Wireshark, Nmap, Cisco
Packet Tracer
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INTRODUCTION

The second week of the Cybersecurity Internship at DG Interns Hub focused on developing practical knowledge of advanced networking, network analysis, and fundamental security concepts. The objective of this week was to understand how devices and services communicate over a network and how network traffic can be monitored, analyzed, and secured.

The assignment covered important concepts including network ports and services, networking devices such as routers, switches, and firewalls, and common network security threats such as Man-in-the-Middle (MITM), Denial-of-Service (DoS), and packet sniffing. Along with theoretical learning, practical exercises were performed using Wireshark, Nmap, and Cisco Packet Tracer. Wireshark was used to capture and analyze network packets, identify DNS requests, examine web traffic, and understand source and destination IP addresses. Nmap was used on an authorized target to perform basic network scanning and identify open ports and running services. Cisco Packet Tracer was used to design and configure a basic LAN and verify connectivity between network devices.

The activities were performed in controlled and authorized environments with a focus on learning, observation, and responsible cybersecurity practices. Overall, this week's work provided a practical understanding of how networking concepts are applied in real-world cybersecurity analysis and network administration.

TASK 1: PORTS & SERVICES

Network ports are logical communication endpoints used by applications and services to communicate over a network. Ports are associated with protocols such as TCP and UDP. Understanding common ports helps in network administration, troubleshooting and security analysis.

Port Number	Protocol	Service Name	Usage
20	TCP	FTP Data	File transfer data
21	TCP	FTP	File transfer protocol
22	TCP	SSH	Secure remote login
23	TCP	Telnet	Remote login
25	TCP	SMTP	Sending emails
53	TCP/UDP	DNS	Domain name resolution
67	UDP	DHCP Server	Assigns IP addresses
68	UDP	DHCP Client	Receives IP configuration
80	TCP	HTTP	Web traffic
110	TCP	POP3	Receiving emails
123	UDP	NTP	Time synchronization

143	TCP	IMAP	Email access
161	UDP	SNMP	Network management
443	TCP	HTTPS	Secure web traffic
445	TCP	SMB	File/printer sharing
465	TCP	SMTPS	Secure email sending
587	TCP	SMTP Submission	Email Submission
993	TCP	IMAPS	Secure email access
995	TCP	POP3S	Secure email receiving
3389	TCP	RDP	Remote desktop access

TASK 2: NETWORKING DEVICES

Router:

A router connects different networks and forwards packets using IP addresses.

Example: Home Wi-Fi router connecting a local network to the Internet.

Switch:

A switch connects devices within the same LAN and forwards frames using MAC addresses.

Example: Connecting computers in a school or office network.

Firewall:

A firewall monitors and filters network traffic according to security rules. It helps block unauthorized access.

Router vs Switch – Difference:

Router: Connects different networks and forwards data using IP addresses.

Switch: Connects devices within the same network and forwards data using MAC addresses.

TASK 3: NETWORK SECURITY BASICS

Network Security:

Network security protects networks, devices, and data from unauthorized access and attacks.

MITM:

An attacker secretly positions themselves between two communicating parties and may intercept or modify communication.

Prevention: HTTPS, VPN, and secure networks.

DoS:

An attacker overwhelms a system or network with excessive requests, making the service unavailable.

Prevention: Firewalls, traffic filtering, and monitoring.

Packet Sniffing:

Packets travelling through a network are captured and analyzed. Unencrypted data may be exposed.

Prevention: HTTPS, encryption, and secure networks.

TASK 4: PACKET ANALYSIS USING WIRESHARK

Objective

To capture and analyze network traffic using Wireshark and identify DNS requests, HTTP/HTTPS traffic, and source/destination IP addresses.

Tools Used:

Wireshark

Procedure:

- Opened Wireshark.
- Selected the active network interface.
- Started packet capture.
- Visited websites such as Google.com.
- Stopped the packet capture.
- Analyzed the captured packets.

Observations

1. DNS Requests:

DNS packets were observed while accessing websites. DNS is used to convert a domain name such as google.com into an IP address.

2. HTTP/HTTPS Traffic:

Web traffic was observed in the captured packets. HTTPS traffic is encrypted and provides secure communication between the browser and web server.

3. Source & Destination IP:

The captured packets showed the Source IP of the device sending the packet and the Destination IP of the device/server receiving it.

Result:

Wireshark was successfully used to capture and analyze network traffic. DNS requests, web traffic, and source/destination IP addresses were identified.

No.	Time	Source	Destination	Protocol	Length	Info
4	0.171329842	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	91	Standard query 0x18a3 AAAA example.com
5	0.17136610	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	91	Standard query 0x8ee6 A example.com
6	0.22266821	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	133	Standard query response 0x7474 HTTPS example.com HTTPS
7	0.22266878	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	119	Standard query response 0x18a3 AAAA example.com AAAA 2696:4790:9645:72db:f264:dsa:ef6b:ff98
8	0.222668878	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	123	Standard query response 0x8ee6 A example.com A 172.66.147.243 A 184.26.23.154
9	0.224673510	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	91	Standard query 0x9a9f HTTPS example.com
10	0.224729564	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	91	Standard query 0x231c AAAA example.com
11	0.224761573	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	91	Standard query 0xe9cf A example.com
12	0.507883216	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	119	Standard query response 0x231c AAAA example.com AAAA 2696:4790:9645:72db:f264:dsa:ef6b:ff98
13	0.507883551	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	123	Standard query response 0xe9cf A example.com A 184.26.23.154 A 172.66.147.243
14	0.648847857	2409:40e3:500f:77a0::	2606:4790:9645:72db::	TCP	94	43614 → 443 [SYN] Seq=654326 Len=0 MSS=1340 SACK_PERM TSval=389863841 TSecr=0 WS=1624
15	0.651233123	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	DNS	133	Standard query response 0xb9f9 HTTPS example.com HTTPS
16	0.651270093	2409:40e3:500f:77a0::	2409:40e3:500f:77a0::	ICMPv6	181	Destination Unreachable (Port unreachable)
17	0.709369679	2606:4790:9645:72db::	2409:40e3:500f:77a0::	TCP	94	443 → 43614 [SYN, ACK] Seq=65535 Len=0 MSS=1300 SACK_PERM TSval=2565716722 TSecr=389863841 WS=6192
18	0.709369757	2409:40e3:500f:77a0::	2606:4790:9645:72db::	TCP	86	43614 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=389863900 TSecr=2565716722
19	0.709369876	2409:40e3:500f:77a0::	2606:4790:9645:72db::	TCP	1374	43614 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=1288 TSval=389863901 TSecr=2565716722 [TCP PDU reassembled in 20]
20	0.709370151	2409:40e3:500f:77a0::	2606:4790:9645:72db::	ICMPv6	589	Client Echo (len=example.com)
21	0.709370159	2606:4790:9645:72db::	2409:40e3:500f:77a0::	TCP	86	443 → 43614 [ACK] Seq=1 Ack=1289 Win=131072 Len=0 TSval=2565716813 TSecr=389863901
22	0.829899460	2606:4790:9645:72db::	2409:40e3:500f:77a0::	TCP	86	443 → 43614 [ACK] Seq=1 Ack=1792 Win=131072 Len=0 TSval=2565716840 TSecr=389863901
23	0.828106331	2606:4790:9645:72db::	2409:40e3:500f:77a0::	ICMPv6	489	[TCP Previous segment not captured] Ignored Unknown Record
24	0.828106912	2606:4790:9645:72db::	2409:40e3:500f:77a0::	TCP	2052	[TCP Retransmission] 443 → 43614 [PSH, ACK] Seq=1 Ack=1792 Win=131072 Len=2576 TSval=2565716849 TSecr=389863901
25	0.828115928	2409:40e3:500f:77a0::	2606:4790:9645:72db::	TCP	98	[TCP Dup ACK 1841] 43614 → 443 [ACK] Seq=1792 Ack=1 Win=64512 Len=0 TSval=389864022 TSecr=2565716848 SLE=3865 SRE=4269
26	0.828206338	2409:40e3:500f:77a0::	2606:4790:9645:72db::	TCP	98	43614 → 443 [ACK] Seq=1792 Ack=2577 Win=78556 Len=0 TSval=389864022 TSecr=2565716849 SLE=3865 SRE=4269
27	0.828206538	2409:40e3:500f:77a0::	2606:4790:9645:72db::	TCP	1374	43614 → 443 [ACK] Seq=1792 Ack=2577 Win=78556 Len=0 TSval=389864022 TSecr=2565716849
Frame 28: Packet, 589 bytes on wire (4712 bits), 589 bytes captured (4712 bits) on interface wlan0, id 0000						
Ethernet II, Src: AzureWaveTec_d9:91:7e (08:41:f4:d9:91:7e), Dst: 62:24:4a:89:b6:b0 (62:24:4a:89:b6:b0)						
Internet Protocol Version 6, Src: 2409:40e3:500f:77a0:f5fa:f404:aa:d9e2, Dst: 2606:4790:9645:72db:f264:						
Transmission Control Protocol, Src Port: 43614, Dst Port: 443, Seq: 1289, Ack: 1, Len: 593						
[2 Reassembled TCP Segments (1701 bytes): #1288, #28(593)]						
Transport Layer Security						
0000 02 24 4a 89 b6 b0 a8 41 f4 d9 91 7e 06 dd 60 0f b53...A...E						
0001 0c 11 02 17 06 40 24 09 40 c3 50 0f 17 05 fa 05 8 p P...						
0002 f4 04 00 aa d3 e2 20 06 47 00 96 45 72 db f2 64 Z k A... (r t						
0003 0d 5a ef 6b ff 98 aa 5e 01 bb c3 70 d3 b5 74 2 k A... (r t						
0004 ee da 08 19 00 3f c8 59 0e 0e 01 91 08 0a 17 3c u ? Y...<						
0005 d9 dd 98 e8 ba f2 ba ff 4c 57 62 d7 c9 ae 53 9e Lwb S						
0006 00 a9 98 33 2c be fd 86 89 62 47 50 2a 55 1c d f...P U						
0007 02 0b c7 67 2d 44 73 b9 e2 16 cd 5a b5 19 7b a8 h j L b...) ?						
0008 7b 19 e7 a9 42 b4 6d f9 95 4f ef 47 56 34 b6 b0 (B m O Q V A						
0009 ae 3b 4e 70 86 7f aa 6c 73 91 52 ad fd 2c 59 10 :np l s R...Y						
000a 00 75 16 27 ab 09 5a e2 ff 33 01 5c bc 47 46 31 u ? S...l F i						
000b 23 6d fc 44 63 b1 11 05 75 1a 8e 14 62 29 57 88 #m Dc...U...b W						
000c 9a 61 53 9d 1c 81 06 88 88 b4 3b 4f 8c 85 4d ca aS...:O M						
000d 00 75 16 27 ab 09 5a e2 ff 33 01 5c bc 47 46 31 u ? S...l F i						
000e 62 1a 3a 5d 98 ab 70 8b e2 c9 22 f0 cf d4 21 b3 b:]...p...H						
000f 00 21 0b 8d 49 17 17 68 31 ff 07 52 99 c4 ba 08 1ik i h 1...R						
0010 0b 4b 43 29 42 1a eb 23 cc 27 c1 44 f7 ba 4d e9 KQ B...#...M						
0011 0120 3c 29 2b 48 36 bc 21 cb c2 d5 13 40 05 5e 6) H6...:...A						
0012 5c 78 c5 da 31 07 35 78 8e 62 53 ad fe 41 2c 01 vx 1 5x...:...A						
0013 00 01 c8 00 b1 24 7f 73 ab c6 14 69 f9 5b 0 w...S...43 p						
0014 21 fe 5a 32 ac f4 15 af 35 0a cf e8 0c c6 08 02 i 22...S...A						
0015 15 02 77 9d ca 40 e2 b9 9d 54 03 bc b3 cc a8 5e w...:...A						
0016 38 78 ad 16 c8 e4 f4 7d 70 3b e2 82 8a 01 5a 8p...:...A						
0017 1a 00 ea cb d3 a1 ab 7f da 00 35 37 8b ca 58 6b57 Xk						

No.	Time	Source	Destination	Protocol	Length	Info
263.7.386664162	10.177.217.197	10.177.217.59	DNS	84	Standard query 0x795d HTTPS www.googleadservices.com	
264.7.386665035	10.177.217.197	10.177.217.59	DNS	84	Standard query 0x99e9 AAAA www.googleadservices.com	
265.7.386708689	10.177.217.197	10.177.217.59	DNS	84	Standard query 0xfeda A www.googleadservices.com	
305.7.429288441	10.177.217.59	10.177.217.197	DNS	189	Standard query response 0x7e9d HTTPS www.googleadservices.com HTTPS	
306.7.429289862	10.177.217.59	10.177.217.197	DNS	196	Standard query response 0x98e9 AAAA www.googleadservices.com AAAA 64:ff9b:c0b2:9e9a AAAA 64:ff9b:c0b2:9e9b AAAA 64:ff9b:c0b2:9e9c	
307.7.429561617	10.177.217.59	10.177.217.197	DNS	148	Standard query response 0xfeda A www.googleadservices.com A 192.178.158.157 A 192.178.158.156 A 192.178.158.154 A 192.178.158.158	
320.7.462771312	10.177.217.197	10.177.217.59	DNS	84	Standard query 0x7e99 HTTPS www.googletagmanager.com	
321.7.462776774	10.177.217.197	10.177.217.59	DNS	84	Standard query 0x71b9 AAAA www.googletagmanager.com	
322.7.462869999	10.177.217.197	10.177.217.59	DNS	84	Standard query 0x1b08 A www.googletagmanager.com	
334.7.462884950	10.177.217.59	10.177.217.197	DNS	141	Standard query response 0x7e99 HTTPS www.googletagmanager.com SOA ns1.google.com	
335.7.506708396	10.177.217.59	10.177.217.197	DNS	112	Standard query response 0x71b9 AAAA www.googletagmanager.com AAAA 2404:6800:4002:819:2000	
336.7.501519318	10.177.217.59	10.177.217.197	DNS	100	Standard query response 0x1b08 A www.googletagmanager.com A 142.250.182.40	
453.0.044974247	10.177.217.197	10.177.217.59	DNS	75	Standard query 0x1928 HTTPS www.gstatic.com	
454.0.045116748	10.177.217.197	10.177.217.59	DNS	75	Standard query 0xa9ef AAAA www.gstatic.com	
455.0.045177069	10.177.217.197	10.177.217.59	DNS	75	Standard query 0x594d A www.gstatic.com	
457.0.048949120	10.177.217.59	10.177.217.197	DNS	103	Standard query response 0xa9ef AAAA www.gstatic.com AAAA 2404:6800:4013:811:5e	
458.0.048949771	10.177.217.59	10.177.217.197	DNS	91	Standard query response 0x594d A www.gstatic.com A 192.178.158.94	
459.0.002959451	10.177.217.59	10.177.217.197	DNS	132	Standard query response 0x1928 HTTPS www.gstatic.com SOA ns1.google.com	
474.0.004604444	10.177.217.197	10.177.217.59	ICMPv6	169	Destination unreachable (Port unreachable)	
1283.0.783055040	10.177.217.197	10.177.217.59	DNS	86	Standard query 0x1e78 HTTPS encrypted-tbns.gstatic.com	
1284.0.783188914	10.177.217.197	10.177.217.59	DNS	86	Standard query 0xe096 AAAA encrypted-tbns.gstatic.com	
1285.0.783278921	10.177.217.197	10.177.217.59	DNS	86	Standard query 0xafef A encrypted-tbns.gstatic.com	
1295.0.816790648	10.177.217.59	10.177.217.197	DNS	143	Standard query response 0x1e78 HTTPS encrypted-tbns.gstatic.com SOA ns1.google.com	
1296.0.816790738	10.177.217.59	10.177.217.197	DNS	190	Standard query response 0xe096 AAAA encrypted-tbns.gstatic.com AAAA 2404:6800:4013:811:5e AAAA 2404:6800:4013:811:5e AAAA 2404:6800:4013:811:5e	
1305.0.819203638	10.177.217.59	10.177.217.197	DNS	102	Standard query response 0xafef A encrypted-tbns.gstatic.com A 142.250.182.238	
Frame 283: Packet, 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface wlan0, id 0						
Ethernet II, Src: AzureWaveTec_09:91:7e (08:41:f4:d9:91:7e), Dst: 62:24:4a:89:b6:b0 (62:24:4a:89:b6:b0)						
Internet Protocol Version 4, Src: 10.177.217.197, Dst: 10.177.217.59						
User Datagram Protocol, Src Port: 52753, Dst Port: 53						
Domain Name System (query)						
0000 02 24 4a 89 b6 b0 a8 41 f4 d9 91 7e 08 00 45 00 b53...A...E						
0001 00 45 00 0c 00 00 40 11 81 67 0a b1 d9 c5 8a b1 F...:...g...						
0002 49 3b ce 11 00 35 00 32 3a 45 7e 9d 01 00 00 01 :...5 2 :E.....						
0003 00 00 00 00 00 00 03 77 77 77 10 67 6f 67 6cw w googl						
0004 45 61 64 73 65 72 76 69 63 65 73 03 63 6f 6d eadervi ces.com						
0005 00 41 00 01A						

TASK 5: NETWORK SCANNING USING NMAP

Objective

To perform basic network scanning and identify open ports and services using Nmap.

Tool Used:

Nmap

Target scanme.nmap.org

Commands Used:

```
nmap scanme.nmap.org nmap -sV  
scanme.nmap.org
```

Procedure:

Opened Command Prompt/Terminal.

Run the basic Nmap scan.

Observed the open ports.

Used -sV to detect service and version information.

Recorded the scan results.

Observations:

Open ports were identified on the target.

Services running on the open ports were detected.

The -sV option provided additional service/version information.

Result:

The Nmap scan was successfully performed on the authorized target, and open ports and services were identified.

```
(kali㉿kali)-[~]
$ sudo nmap -sV scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-14 12:08 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
9929/tcp   open      nping-echo   Nping echo
31337/tcp  open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.43 seconds
```

Figure 1: Nmap service and version detection scan

```
(kali㉿kali)-[~]
$ sudo nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-14 12:10 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
25/tcp    filtered  smtp
80/tcp    open      http
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
9929/tcp   open      nping-echo
31337/tcp  open      Elite

Nmap done: 1 IP address (1 host up) scanned in 12.38 seconds
```

Figure 2: Nmap basic network scan

TASK 6: PACKET TRACER LAN SETUP

Objective

To create a simple LAN using two PCs and one switch and test connectivity between the PCs.

Tool Used

Cisco Packet Tracer

Topology

PC1 → Switch ← PC2

IP Configuration

PC1: 192.168.1.10

PC2: 192.168.1.11

Procedure

- Created 2 PCs and 1 Switch in Cisco Packet Tracer.
- Connected both PCs to the switch.
- Configured the IP address of PC1 as 192.168.1.10.
- Configured the IP address of PC2 as 192.168.1.11.
- Used the Ping command to test connectivity.

Result

The ping was successful, confirming that PC1 and PC2 were able to communicate through the switch.

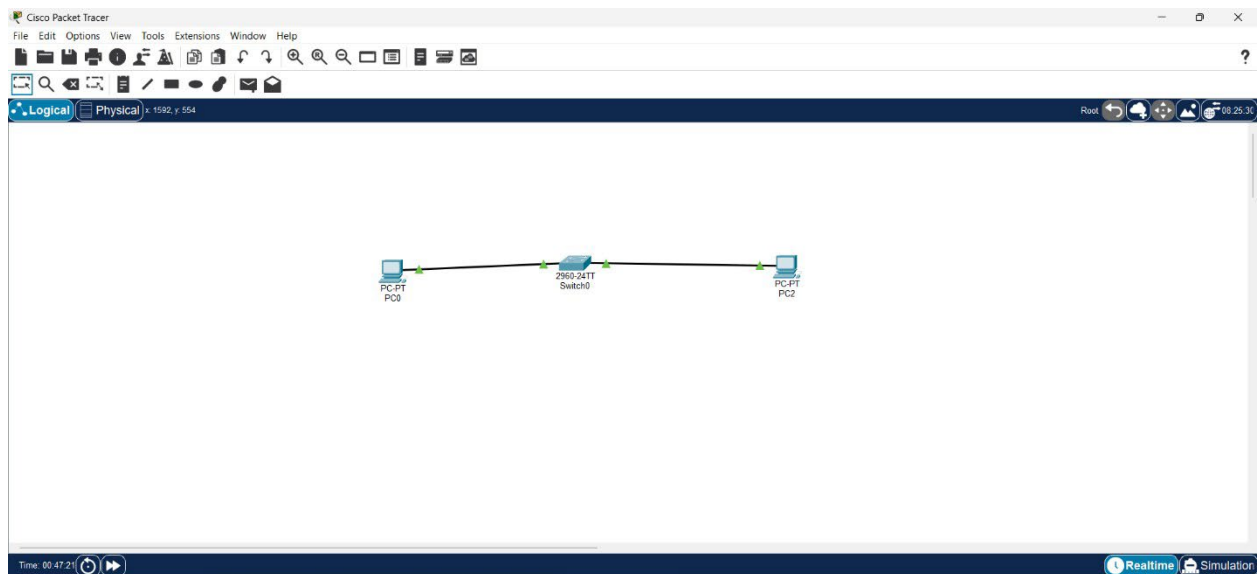


Figure 1: Two PCs connected through a switch

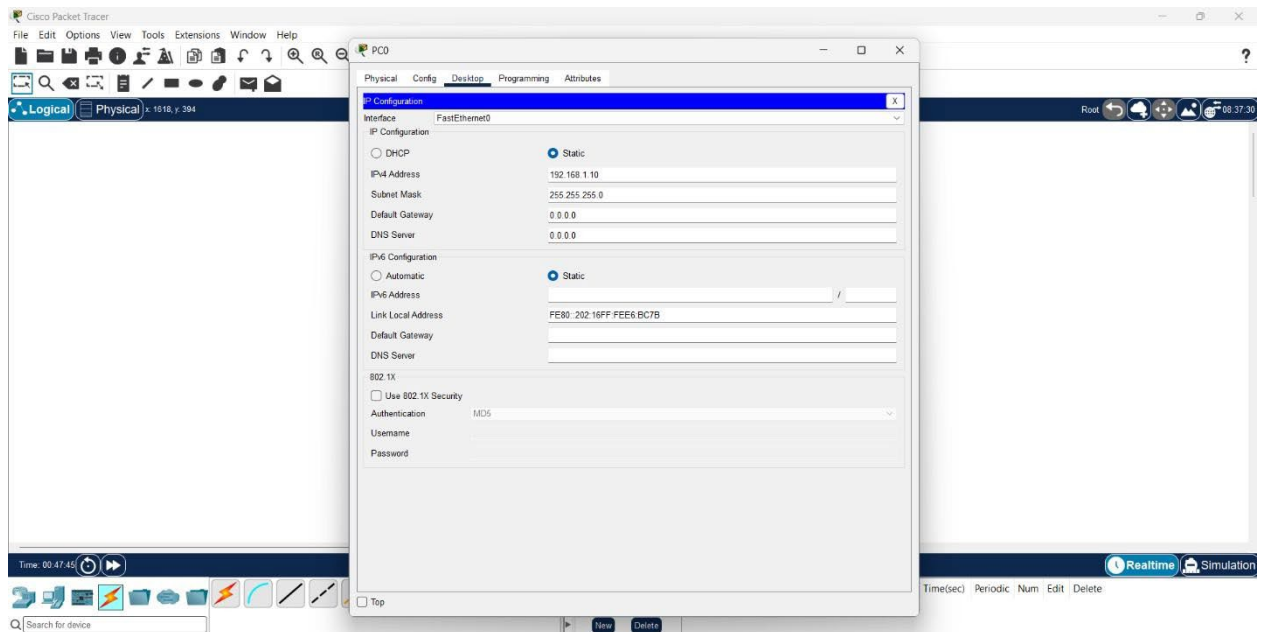


Figure 2: IP address configuration

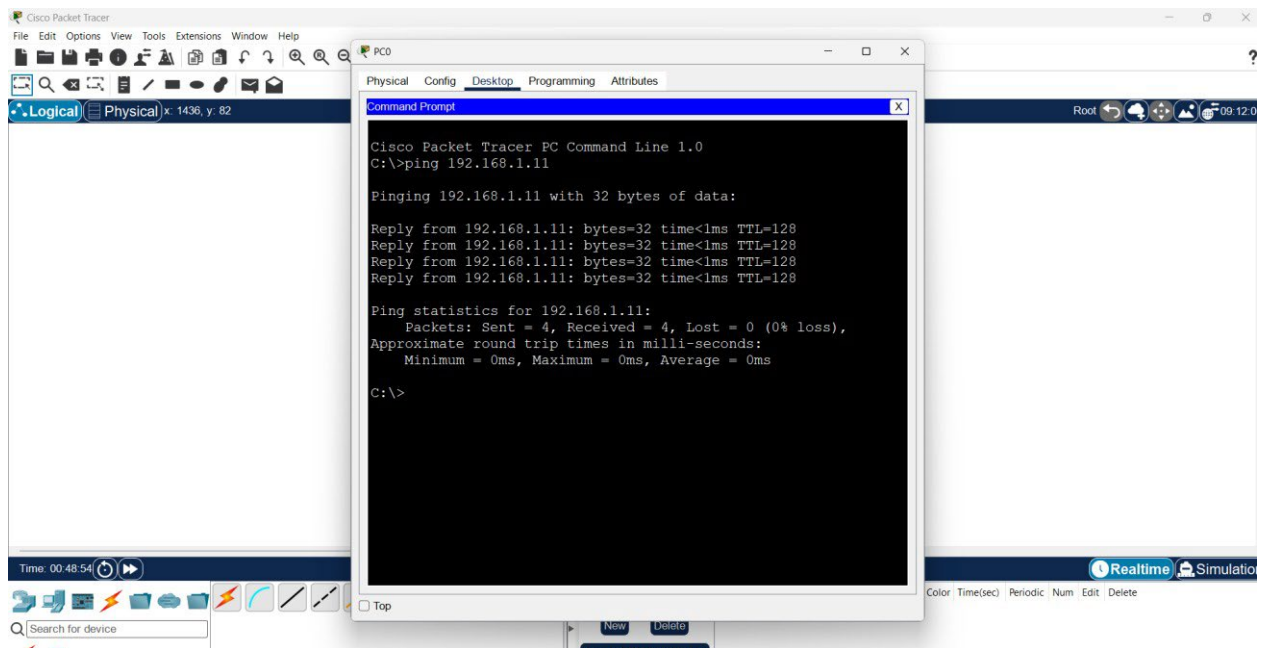


Figure 3: Ping test successfully

TASK 7: MITM CONCEPT- SAFE SIMULATION Objective:

To understand the concept and risks of a Man-in-the-Middle (MITM) attack in a safe environment.

Topology:

PC1(Victim)-> Switch -> Router

PC2(Attacker)-> Switch

IP Configuration:

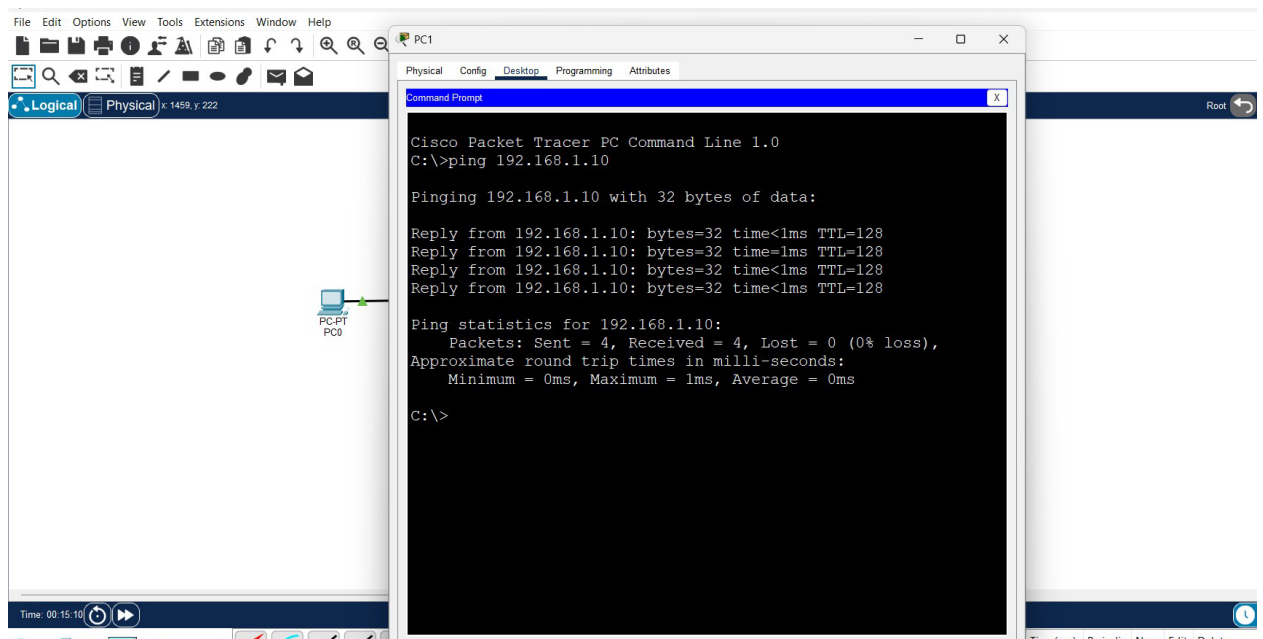
PC1: 192.168.1.10

PC2: 192.168.1.20

Router: 192.168.1.1

Procedure:

- Created PC1, PC2, Switch and Router.
- Connected the devices using the appropriate cables.
- Assigned the specified IP addresses.
- Tested normal network connectivity using ping.
- Studied how an attacker could position themselves between communicating devices.



CONCLUSION

- Developed a clear understanding of **common network ports, protocols, and services** and their practical applications.
- Learned the functions and differences between major networking devices, including **routers, switches, and firewalls**.
- Studied fundamental network security threats such as **MITM attacks, DoS attacks, and packet sniffing**, along with their basic prevention techniques.
- Gained practical experience using **Wireshark** to capture and analyze network packets, DNS requests, web traffic, and IP addresses.
- Used **Nmap** in an authorized environment to perform basic network scanning and identify open ports, services, and service-version information.
- Created and configured a basic **LAN topology using Cisco Packet Tracer** and successfully verified connectivity between PCs using the ping command.
- Improved understanding of how attackers may attempt to position themselves within network communication through a **safe MITM concept simulation**, without performing an actual attack.
- Overall, the week strengthened my **practical networking and cybersecurity fundamentals** and provided a foundation for further learning in network security, vulnerability assessment, and ethical security testing.